Data Management Project

Building a Comprehensive Music Dataset for Sentiment Analysis

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OUR APPROACH TO DATA LIFECYCLE

Managing and Integrating Music Data

Research question

Identify a problem and form a thesis statement.

Data Acquisition

Identifying the data sources and choose the method to acquire the data

Data Integration

Integrating data from both same data source and different data sources

Data Cleaning

Handling with conflicts and heterogeinities

Data Quality

Choosing the dimensions to evaluate the final dataset obtained

RESEARCH QUESTIONS

What makes a song a global hit?

Are there any relations between the lyric and the sound?

How a song should be written to catch the public's attention?

Can a good text be replicated by an AI?

DATA ACQUISITION

SPOTIFY

- Method: Accessed via Spotify API.
- Playlist: Top 50 Global Songs playlists for 2020, 2021, 2022, and 2023.
- Variables Collected: ID, Song title, Artist(s), Album, Release date, Popularity score

KAGGLE

- Method: Download
- Playlist: Top 50 Global Songs playlists for 2020, 2021, 2022, and 2023.
- Variables Collected: Song title, Artist(s), Album, Release date, Energy, Danceability, Acousticness, Valence, Tempo, Loudness

AZLyrics

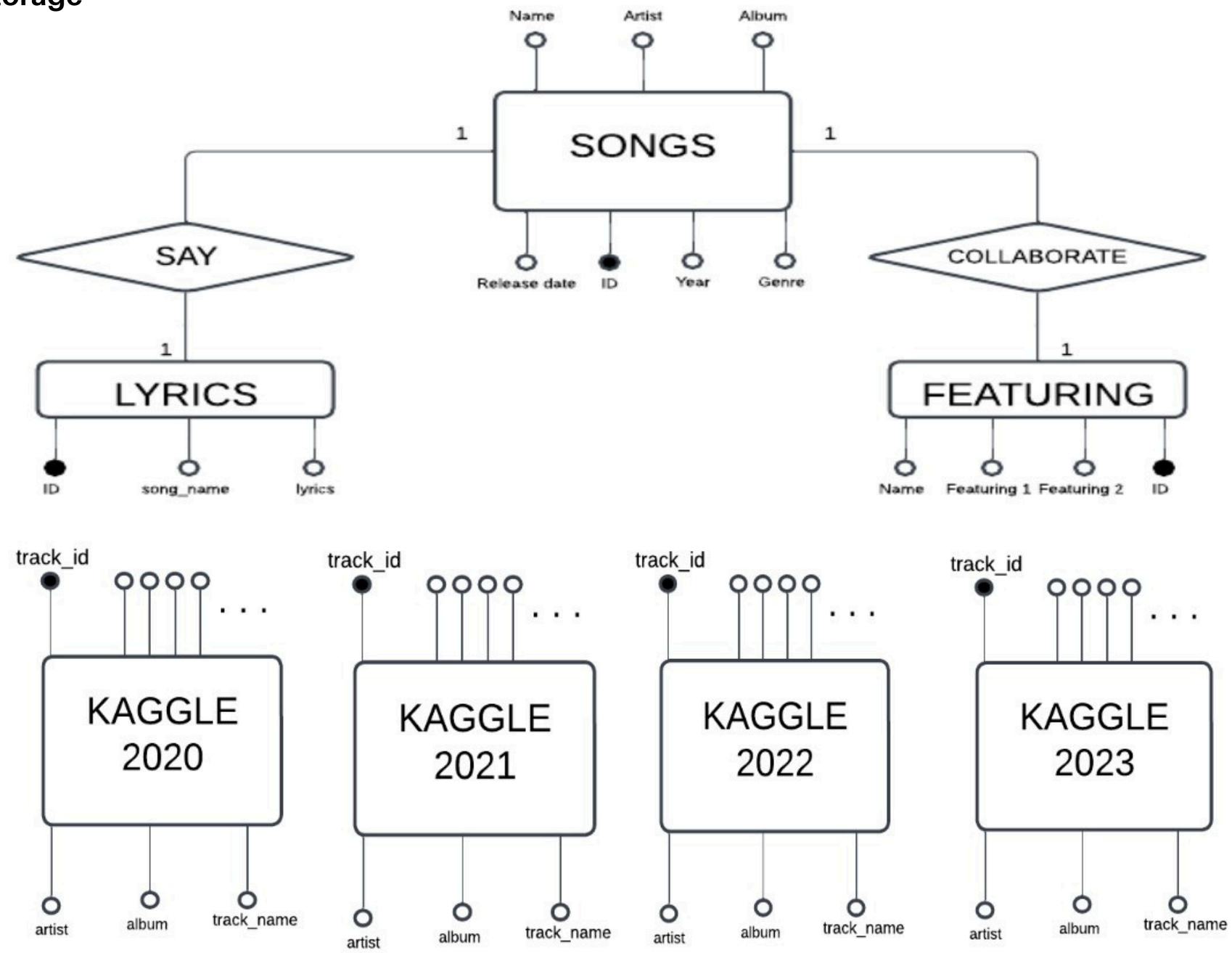
- Method: Accessed via Web Scraping
- Generate URL, Clean Input, Handle Featuring Artists, Name Mapping

DATA STORAGE

Why the Relational Model?

FIXED TABLE STRUCTURES	PREDICTABLE UPDATES	LIGHTWEIGHT JOINS	STRUCTURED DATA TYPES
Well-defined schema for seamless and efficient updates/extensions.	Annual influx of top songs ensures stability and compatibility with relational databases.	Simple relationships between tables allow efficient queries even as the dataset grows.	Majority of variables (e.g., track attributes, metadata) are structured, making them ideal for relational modeling.

E-R model after data storage



DATA INTEGRATION

OBJECTIVE

Integrate the four years datasets downloaded from Kaggle

1

SOLUTION

Selected key features shared across all Kaggle datasets and adjustments for name heterogeneity to directly concatenate.

OBJECTIVE

Integrate the Spotify and the final Kaggle dataset into a unified and unique dataset

2

SOLUTION

Merged datasets using: Track ID, Track Name, and Artist Name as keys. Resulted in ~100 songs in common between Spotify and Kaggle datasets.

OBJECTIVE

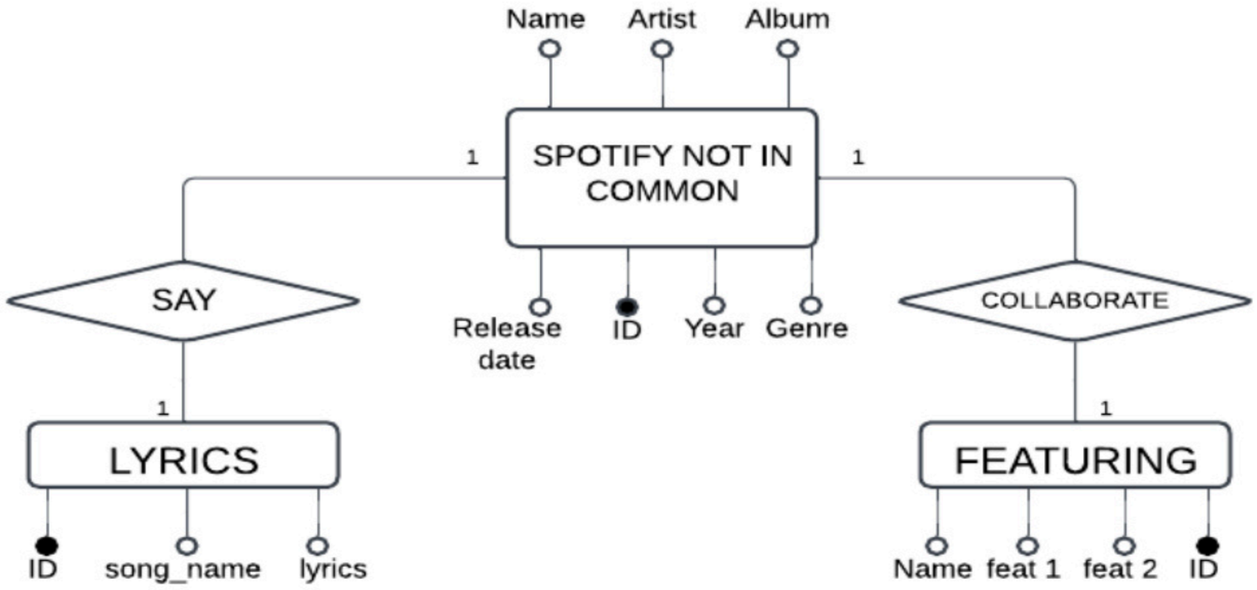
Using web scraping to add the column 'lyrics' to the final dataset.

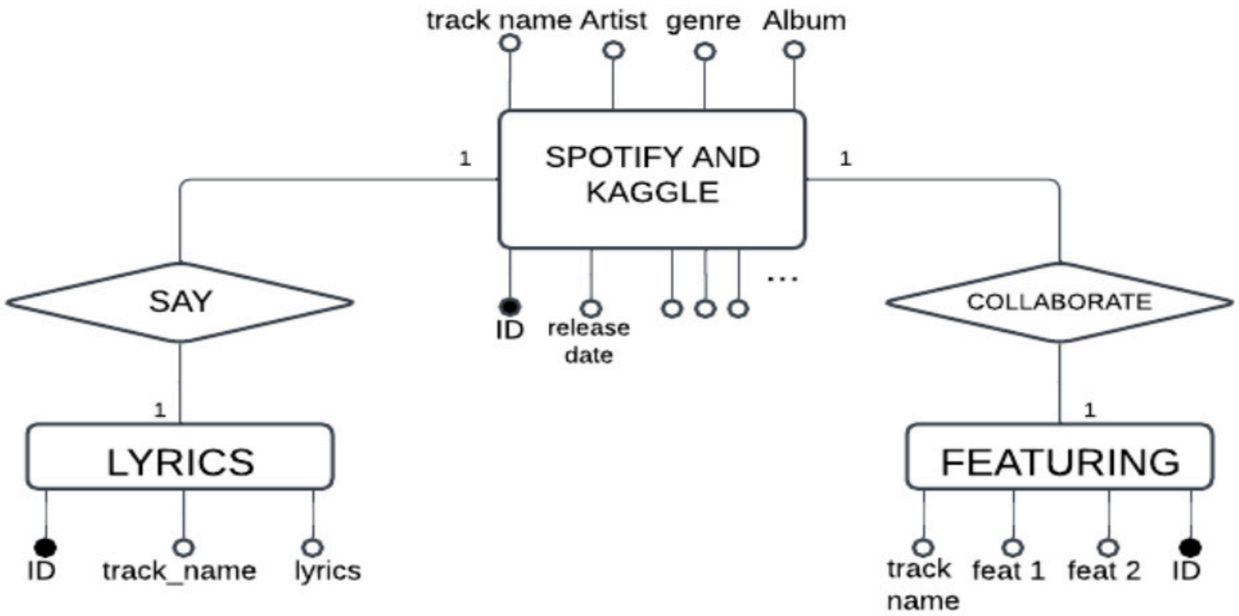
3

SOLUTION

Created a structured foundation for integrating the lyrics data in the next step.

E-R model after data integration





DATA CLEANING

ADJUSTING SONG TITLES	Removed information about featured artists from song titles. Stored the featuring data in a separate Featuring Table.
HANDLING DUPLICATES	Identified and removed duplicate records while adhering to the Minimization Principle, ensuring no redundant data.
RAW LYRICS	Left lyrics unprocessed (no lemmatization or tokenization) to preserve their original format, allowing flexibility for future analyses.
GENRE GRANULARITY	Retained all genres and sub-genres without clustering for maximum detail.

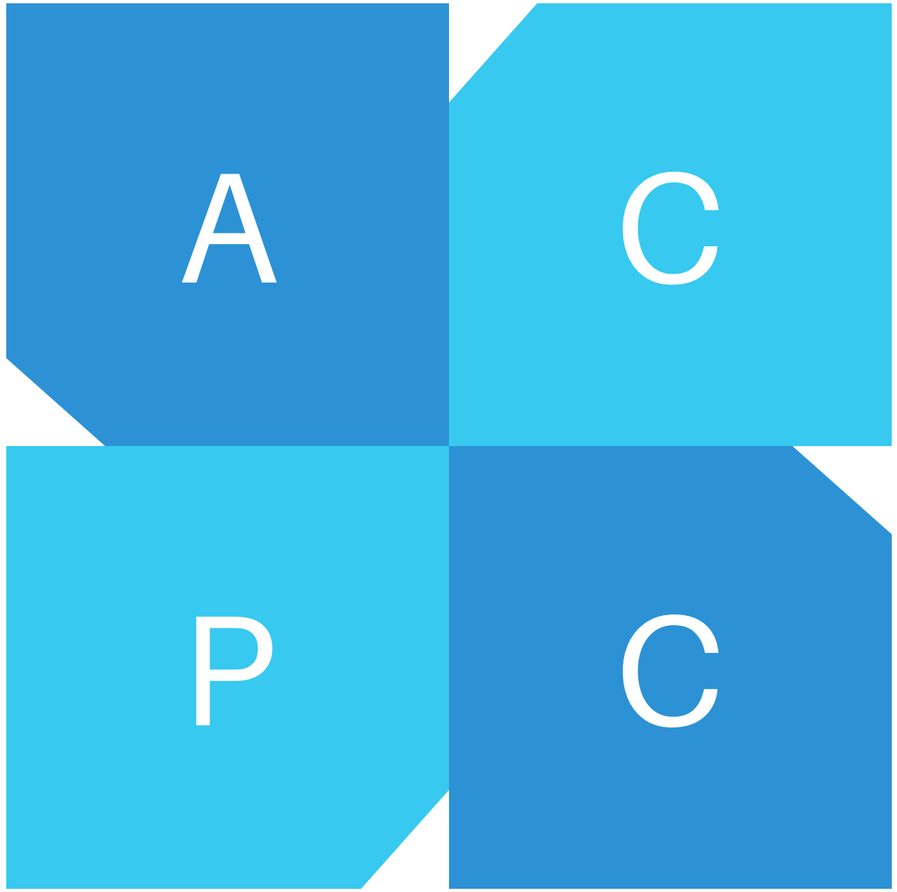
DATA QUALITY

ACCURACY

All the informations we stored and integrated had to be accurate and reflect the real features and lyrics.

PERTINENCE

The choice to keep only the official top 50 songs for each year was necessary to be coherent with the goals we decided.



CONSISTENCY

A deduplication method was implemented to eliminate duplicates records.

COMPLETENESS

A table completeness formula was implemented to handle missing value.

POSSIBLE DATA EXPLORATION

BASIC EDA: VIOLIN PLOTS, TIME SERIES, CORRELATIONS

SENTIMENTAL ANALYSIS AND INTRODUCTION OF 'MOOD'

NUMBER OF WORDS PER SONG, PER GENRE OR RATIO WITH THE DURATION

TRAIN NLP MODEL TO CREATE NEW LYRICS

CONCLUSION

- This project highlights the power of integrating diverse data sources to create a comprehensive dataset for the music industry. By combining Spotify metadata, Kaggle audio features, and AZLyrics content, we built a resource ready for advanced analysis.

- Using a relational database ensured data consistency and scalability, overcoming challenges like name heterogeneity and missing identifiers. While focused on data preparation, this work sets the stage for future studies, including sentiment analysis, trend exploration, and audience behavior modeling.

- It emphasizes the importance of robust data management for meaningful insights and innovation in the music industry.

THANK YOU!